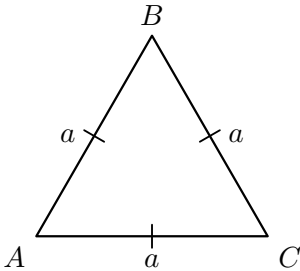
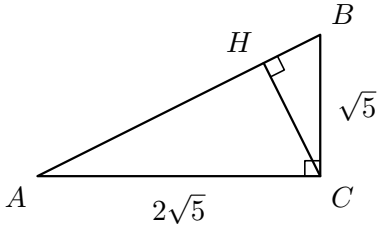
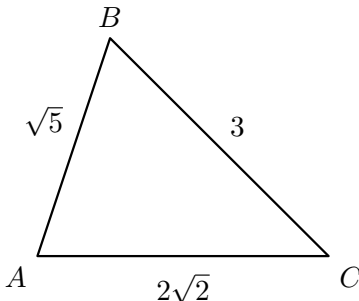
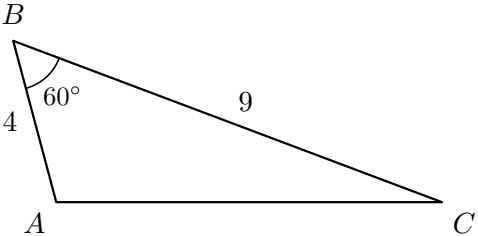
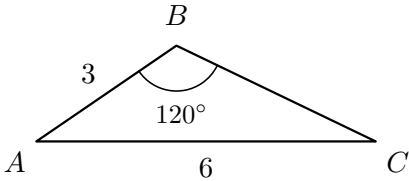
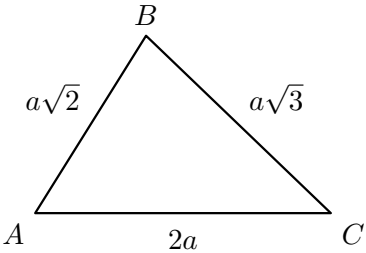
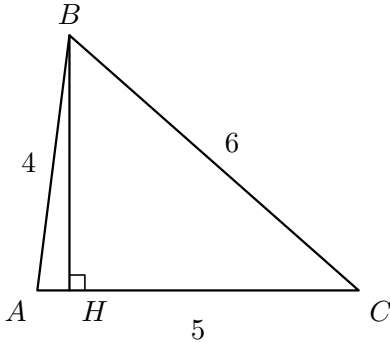
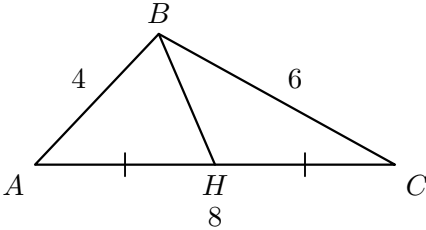
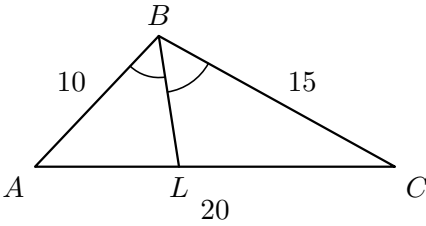
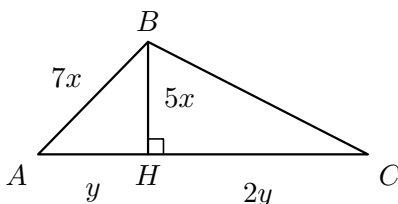
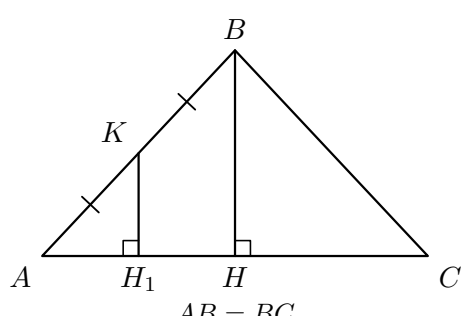
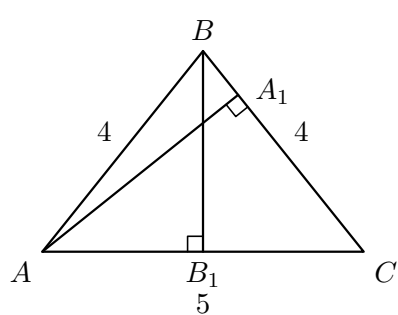
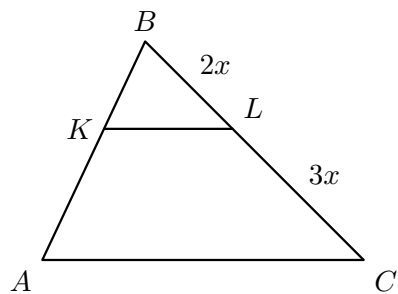
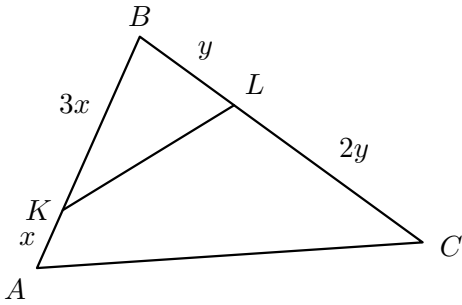
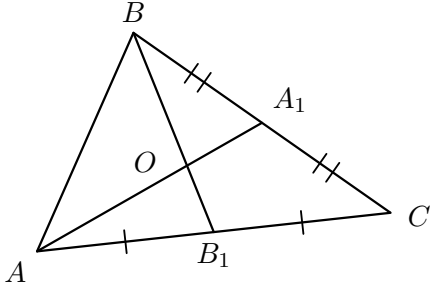
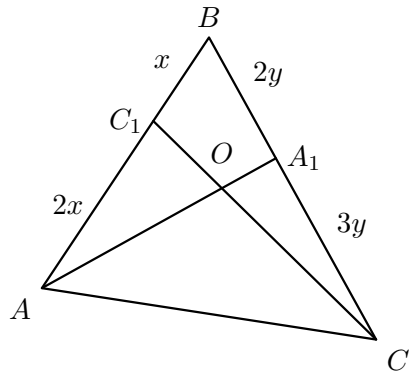
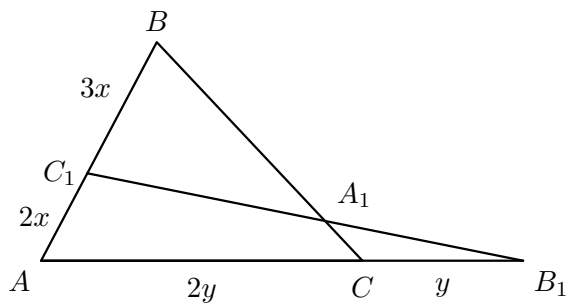
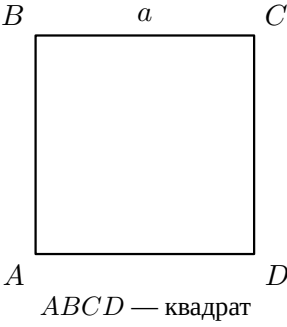


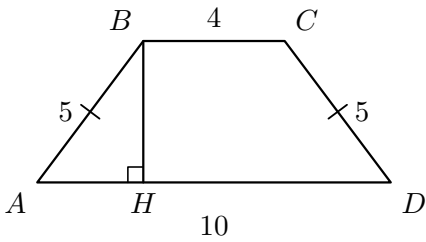
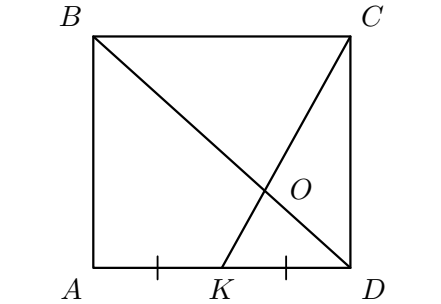
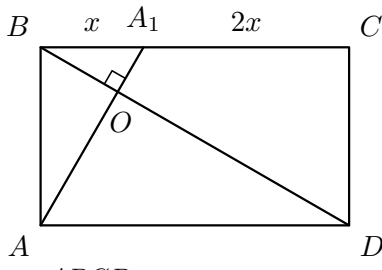
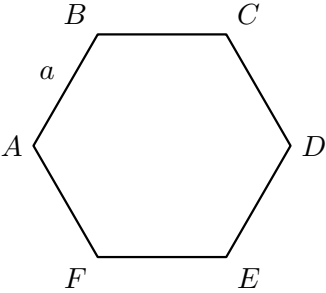
Опорные задачи планиметрии

№	Задание	Ответы
1		$h =$ $S_{ABC} =$ $R_{\text{оп}} =$ $r_{\text{вп}} =$
2		$AB =$ $CH =$ $AH =$ $BH =$ $R_{\text{оп}} =$ $r_{\text{вп}} =$ $\sin \angle BAC =$
3		$S_{ABC} =$
4		$AC =$ $S_{ABC} =$

№	Задание	Ответы
5		$BC =$ $\sin \angle BCA =$
6		$\cos \angle ABC =$
7		$BH =$ $S_{ABC} =$ $AH =$ $HC =$
8		$BH =$
9		$BL =$ $AL =$ $LC =$ $\frac{S_{ABL}}{S_{BLC}} =$ $\frac{S_{BLC}}{S_{ABC}} =$

№	Задание	Ответы
10		$\operatorname{tg} \angle BCA =$
11	 $AB = BC$	$\frac{AH_1}{H_1H} =$ $\frac{AH_1}{AC} =$
12		$\cos \angle BCA =$ $\operatorname{tg} \angle BCA =$ $\cos \angle ABC =$ $BB_1 =$ $AA_1 =$ $R_{\text{оп}} =$ $r_{\text{вп}} =$ $\rho(A_1; AC) =$ $\rho(B_1; BC) =$ $P_{ABC} =$
13	 $KL \parallel AC$	$\frac{BK}{KA} =$ $\frac{KA}{BA} =$

№	Задание	Ответы
14		$\frac{S_{BKL}}{S_{ABC}} =$ $\frac{S_{BKL}}{S_{AKLC}} =$
15		$\frac{BO}{OB_1} =$ $\frac{OB_1}{BB_1} =$
16		$\frac{CO}{OC_1} =$ $\frac{AO}{OA_1} =$
17		$\frac{BA_1}{A_1C} =$ $\frac{A_1C}{BC} =$
18	 $ABCD$ — квадрат	$BD =$ $R_{\text{оп}} =$ $r_{\text{вп}} =$ $S_{ABCD} =$

№	Задание	Ответы
19	 $ABCD$ — равнобедренная трапеция	$AH =$ $BH =$ $S_{ABCD} =$
20	 $ABCD$ — прямоугольник, $AK = KD$	$\frac{DO}{OB} =$ $\frac{KO}{KC} =$
21	 $ABCD$ — прямоугольник	$\frac{AB}{BC} =$
22	 $ABCDEF$ — правильный шестиугольник	$\angle BAF =$ $\angle BAC =$ $\angle BAD =$ $BE =$ $BF =$ $S_{ABCDEF} =$ $R_{\text{оп}} =$ $r_{\text{вп}} =$

Ответы

1. $h = \frac{a\sqrt{3}}{2}$; $S_{ABC} = \frac{a^2\sqrt{3}}{4}$; $R_{\text{оп}} = \frac{a\sqrt{3}}{3}$; $r_{\text{вп}} = \frac{a\sqrt{3}}{6}$
2. $AB = 5$; $CH = 2$; $AH = 4$; $BH = 1$; $R_{\text{оп}} = 2,5$; $r_{\text{вп}} = \frac{3\sqrt{5}-5}{2}$; $\sin \angle BAC = \frac{\sqrt{5}}{5}$
3. $S_{ABC} = 3$
4. $AC = \sqrt{61}$; $S_{ABC} = 9\sqrt{3}$
5. $BC = \frac{-3+3\sqrt{13}}{2}$; $\sin \angle BCA = \frac{\sqrt{3}}{4}$
6. $\cos \angle ABC = \frac{\sqrt{6}}{12}$
7. $BH = \frac{3\sqrt{7}}{2}$; $S_{ABC} = \frac{15\sqrt{7}}{4}$; $AH = 0,5$; $HC = 4,5$
8. $BH = \sqrt{10}$
9. $BL = 3\sqrt{6}$; $AL = 8$; $LC = 12$; $\frac{S_{ABL}}{S_{BLC}} = \frac{2}{3}$; $\frac{S_{BLC}}{S_{ABC}} = \frac{3}{5}$
10. $\operatorname{tg} \angle BCA = \frac{5\sqrt{6}}{24}$
11. $\frac{AH_1}{H_1H} = 1$; $\frac{AH_1}{AC} = \frac{1}{4}$
12. $\cos \angle BCA = \frac{5}{8}$; $\operatorname{tg} \angle BCA = \frac{\sqrt{39}}{5}$; $\cos \angle ABC = \frac{7}{32}$; $BB_1 = \frac{\sqrt{39}}{2}$; $AA_1 = \frac{5\sqrt{39}}{8}$; $R_{\text{оп}} = \frac{16\sqrt{39}}{39}$; $r_{\text{вп}} = \frac{5\sqrt{39}}{26}$; $\rho(A_1; AC) = \frac{25\sqrt{39}}{64}$; $\rho(B_1; BC) = \frac{5\sqrt{39}}{16}$; $P_{ABC} = 13$
13. $\frac{BK}{KA} = \frac{2}{3}$; $\frac{KA}{BA} = \frac{3}{5}$
14. $\frac{S_{BKL}}{S_{ABC}} = \frac{1}{4}$; $\frac{S_{BKL}}{S_{AKLC}} = \frac{1}{3}$
15. $\frac{BO}{OB_1} = 2$; $\frac{OB_1}{BB_1} = \frac{1}{3}$
16. $\frac{CO}{OC_1} = \frac{9}{4}$; $\frac{AO}{OA_1} = \frac{10}{3}$
17. $\frac{BA_1}{A_1C} = \frac{9}{2}$; $\frac{A_1C}{BC} = \frac{2}{11}$
18. $BD = a\sqrt{2}$; $R_{\text{оп}} = \frac{a\sqrt{2}}{2}$; $r_{\text{вп}} = \frac{a}{2}$; $S_{ABCD} = a^2$
19. $AH = 3$; $BH = 4$; $S_{ABCD} = 28$
20. $\frac{DO}{OB} = \frac{1}{2}$; $\frac{KO}{KC} = \frac{1}{3}$
21. $\frac{AB}{BC} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
22. $\angle BAF = 120^\circ$; $\angle BAC = 30^\circ$; $\angle BAD = 60^\circ$; $BE = 2a$; $BF = a\sqrt{3}$; $S_{ABCDEF} = \frac{3\sqrt{3}a^2}{2}$; $R_{\text{оп}} = a$; $r_{\text{вп}} = \frac{a\sqrt{3}}{2}$